

# Daniel Thédié

✉ [daniel.thedie@ed.ac.uk](mailto:daniel.thedie@ed.ac.uk) 🌐 [Website](#) 🐙 [GitHub](#) 🆔 [ORCID](#)

Research Software Engineer with 10 years of experience in academic research and data management. Maintainer of an online database serving 2,000+ registered users internationally, for which I rapidly adopted the Angular and Spring Boot frameworks. Highly experienced in software engineering best practices, data management and curation. Proficient in multiple programming languages and confident in my ability to master new languages quickly and independently.

## Skills

---

|                        |                                        |
|------------------------|----------------------------------------|
| <b>Programming</b>     | Python, Java, R, Matlab, SQL           |
| <b>Web</b>             | HTML, CSS, JavaScript, TypeScript      |
| <b>Frameworks</b>      | Angular, Spring Boot, Bootstrap, Astro |
| <b>Developer tools</b> | Git, Unix, Linux, Docker               |

## Education and Employment

---

**Research Software Engineer**  
*University of Edinburgh, Scotland*

2025–present

- Writing production-grade code
  - Unit testing
  - Continuous Integration pipelines (GitHub Actions)
  - Accessibility testing (Web Content Accessibility Guidelines)
  - Docker containers for reproducible environments
- Maintenance and development of the **BioDare2** data repository and analysis platform
  - Full-stack development following MVC architecture (Angular front-end, Spring Boot back-end, REST API)
  - Responsive web design for cross-device compatibility
  - MySQL database management
- Development of a data management system for a clinical trial (department of Psychiatry)
  - Automated data ingestion to a data store via Extract-Transform-Load pipelines
  - Iterative, user-driven development with regular feedback cycles
  - Interface design and styling using the Bootstrap CSS framework
  - Documentation and user support
  - Work management through GitHub Issues and Projects
- Research funding experience
  - Submitted several grant applications as principal and co-investigator
  - Reviewed grant applications

**Post-doctoral research associate**  
*University of Edinburgh, Scotland*

2020–2024

- *In-vivo* dynamics of the RecBCD repair complex upon exposure of *E. coli* to antibiotics
  - Advanced fluorescence imaging in living cells
  - Automated image acquisition, management and analysis of large datasets
- Development of a Python package (**PyBerries**) for data analysis in the lab
  - Produced stable, reliable code with unit testing and continuous integration
  - Distributed package and provided updates to multiple users
- Maintenance and supervision
  - Regular microscope maintenance
  - Training for new users
  - Supervision of Master students

## Ph.D. Student

2016–2019

*Institute of Structural Biology (IBS), Grenoble, France*

Characterisation of photoconvertible fluorescent proteins for advanced single-molecule localisation microscopy

- Data analysis and simulation (Matlab)
- Communicating results through conference presentations, peer-reviewed articles and reports
- Independent and team work with international collaborators (Belgium, Germany)

## Biotechnology engineering student

2013–2016

*Ecole Supérieure de Biotechnologie de Strasbourg (ESBS), France*

Awarded an Engineering degree in Biotechnology and a Master's degree in structural biology

- Multi-lingual curriculum (English, French, German)
- Advanced training in varied disciplines including genetic engineering, programming (Python, R, Matlab), chemistry, mathematics, statistics
- 4 internships (2 to 6 months, France and Germany)

## Bachelor student in biology

2010–2013

*Cergy-Pontoise University, France*

Ranked first out of 131 students

- Core training in cellular and molecular biology
- Erasmus year at Copenhagen University, Denmark
- 2 internships (1 and 3 months, France and Denmark)

## Software

---

### Active Development

**BioDare2** (Angular, TypeScript, Java)

Maintainer since 2025. Data repository and analysis platform for circadian data ([biodare2.ed.ac.uk](https://biodare2.ed.ac.uk))

**Ambient downloader** (Python)

ETL pipeline to extract and manage sleep data from radar devices via API

**REDCap downloader** (Python)

ETL pipeline to extract and format patient data from the REDCap platform via API

**nocturn** (R, R Shiny, Bootstrap)

Process and visualise sleep data. Includes an **R Shiny app** (available on [CRAN](#))

**svgedit** (R)

Create reproducible, complex figure layouts by inserting elements into svg files (available on [CRAN](#))

### Past projects

**PyBerries** (Python)

Analyse quantitative data from microscopy images

## Teaching

---

### Workshop on git and GitHub (0.5 day)

2022, 2026

*Internal*

Basics of using git for code management, good coding practices ([workshop materials](#)).

|                                                                                                                                      |            |
|--------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Data anonymisation workshop (0.5 day)</b><br><i>Edinburgh Circadian Mental Health Conference</i>                                  | 2025       |
| Introduction to data anonymisation for researchers from different disciplines and career stages.                                     |            |
| <b>Dealing with Data - PhD induction (1 day)</b><br><i>University of Edinburgh</i>                                                   | 2025       |
| Introduction to data management best practices for first year PhD students in the School of Biological Sciences.                     |            |
| <b>Workshop on the BACMMAN Fiji plugin (2 days)</b><br><i>Internal lab training</i>                                                  | 2024       |
| Performing image analysis tasks (segmentation, image quantification) based on the participants' own data, troubleshooting            |            |
| <b>Teaching assistant, cell biology practicals (2 weeks)</b><br><i>Grenoble-Alpes University</i>                                     | 2018, 2019 |
| Helped students conceive experimental projects around mammalian cell culture and fluorescence microscopy. Marked oral presentations. |            |
| <b>Supervision of Masters students</b>                                                                                               |            |
| Supervised 4 Master students between 2018 and 2023 on microscopy and programming projects.                                           |            |

## Academic experience

---

8 peer-reviewed publications (2017 - 2025)

- [Full publication list](#)

Attended 15 conferences

- 6 oral presentations, 7 posters

## Outreach activities

---

|                                                                          |           |
|--------------------------------------------------------------------------|-----------|
| <b>Edinburgh Science Festival</b>                                        | 2023      |
| Public engagement on microscopy, bacteria and antimicrobial resistance   |           |
| <b>Member of a young scientists committee</b><br><i>Grenoble, France</i> | 2018-2019 |
| Organising research seminars                                             |           |